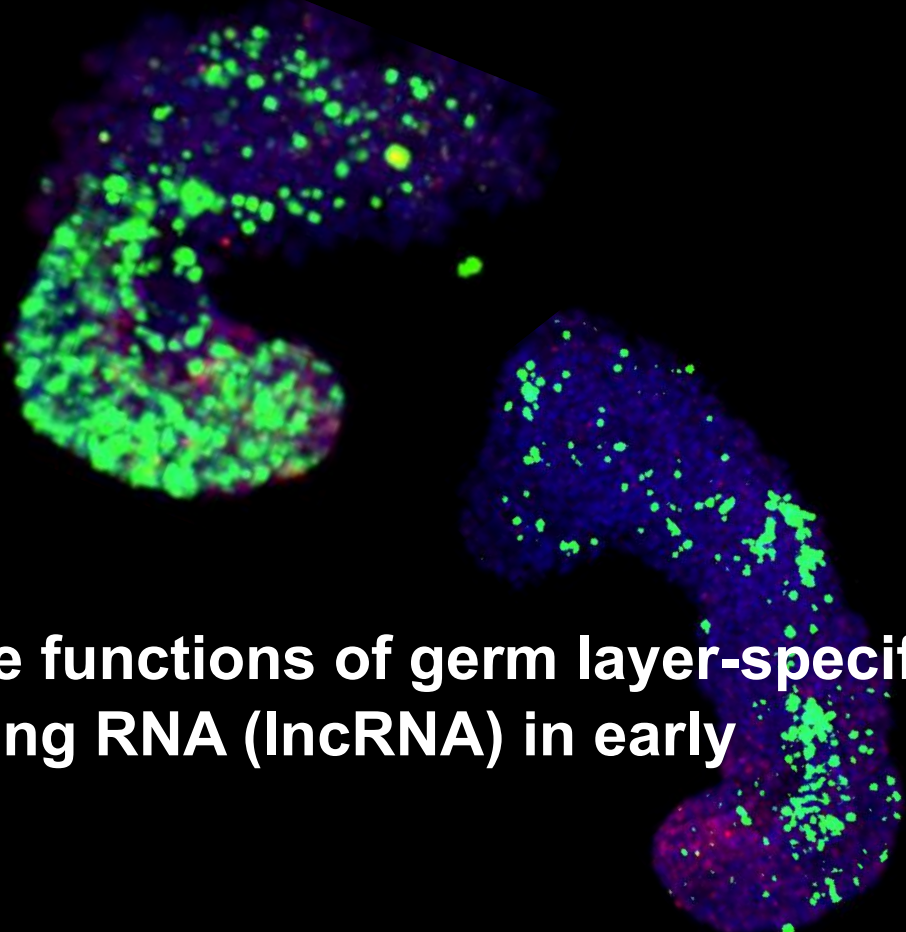




Unraveling the functions of germ layer-specific long non-coding RNA (lncRNA) in early development

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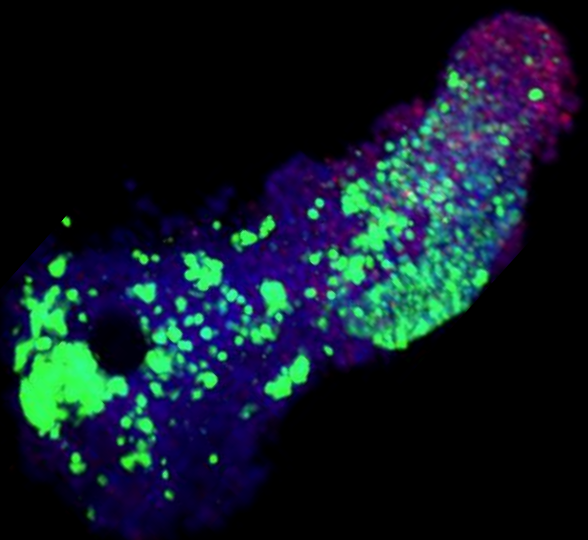
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Examiner: Dr. Bernard Roelen



Cover figure: Gastruloids generated from Sox2::H2B-Venus (green) and Brachyury (T) ::H2B-mCherry (red) mouse embryonic stem cells (mESCs) at day 5 of differentiation. Nuclear staining with DAPI (blue) highlights cell nuclei. Images were captured using a Dragonfly 500 microscope at 10X magnification.

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Layman's Summary

During early embryonic development, stem cells differentiate into three primary layers: ectoderm, mesoderm, and endoderm. The ectoderm (outer layer) gives rise to the nervous system, which includes the brain and spinal cord, as well as sensory organs like the eyes and ears; the mesoderm (middle layer), which gives rise to muscles, bones, and the cardiovascular system, including the heart and blood vessels; and the endoderm (innermost layer), which forms essential organs like the liver, lungs, and the digestive system. If any of these layers fail to develop correctly, it can lead to embryo loss or developmental disorders. To prevent such outcomes, researchers work to understand the molecular “instructions” that guide this process.

Previously, scientists focused on protein-coding RNAs, which direct protein synthesis and were believed to be the primary regulators of cell development. However, recent studies highlight the role of long non-coding RNAs (lncRNAs), which regulate cellular processes without producing proteins. These lncRNAs interact with DNA, RNA, or proteins to control gene expression. Advances in single-cell RNA sequencing have identified numerous lncRNAs in embryonic development, yet many of their functions remain unknown. Our study examines three lncRNAs—RMST, C13Rik, and 913Rik—which exhibit specific expression patterns in early embryonic cells. RMST and C13Rik are primarily found in the ectoderm, while 913Rik is localized in the mesoderm.

To investigate their role, we employed two model systems: 2D differentiation cell cultures and 3D gastruloids. In the 2D system, we optimized protocols to direct mouse stem cells into forming ectoderm or mesoderm tissues. In the 3D system, we developed gastruloids, which are small cell clusters that mimic early embryonic development.

Using gene knockdown techniques, we ‘reduced’ the activity of RMST, C13Rik, or 913Rik in these models to observe their effect on differentiation. Our preliminary findings suggest that RMST, C13Rik, and 913Rik play roles in directing stem cell fate. Gaining a deeper understanding of these lncRNAs will enhance our knowledge of embryonic development and may provide insights into developmental disorders (Figure 1).

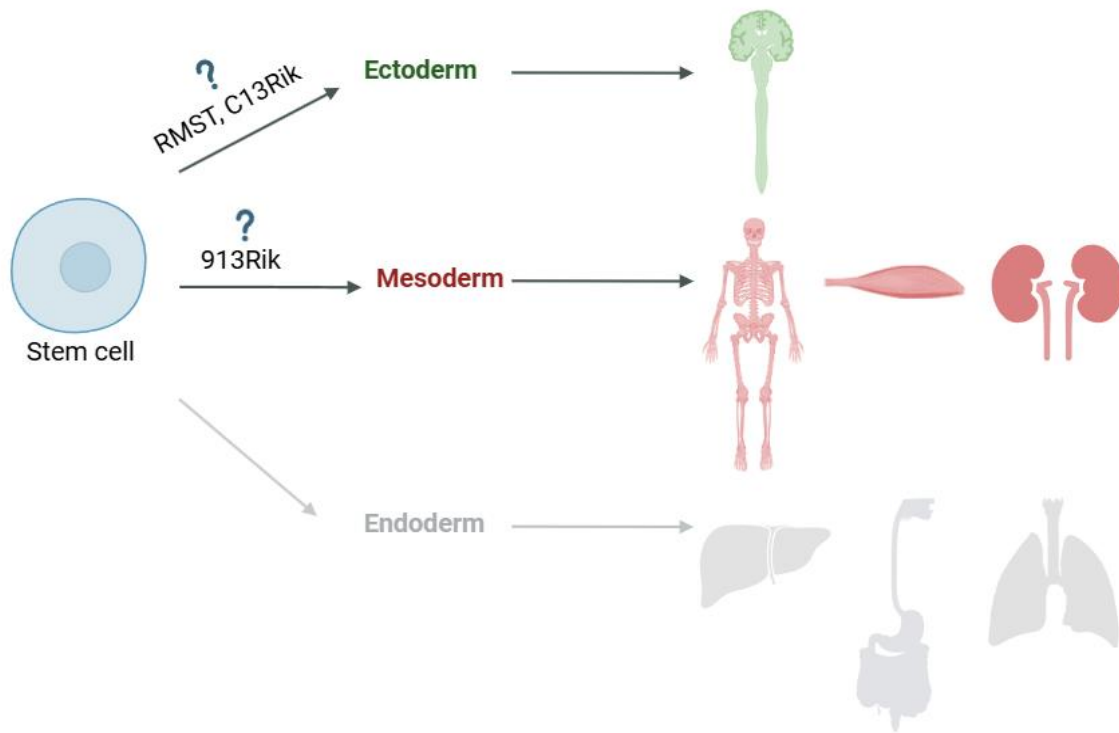


Figure 1 | Layman's graphical abstract. RMST, C13Rik, and 913Rik are long non-coding RNAs studied in this research for their potential role in guiding cell fate during embryonic development. The figure was created in BioRender.com.

Abstract

Background: Cell fate commitment during early embryonic development directs pluripotent stem cells into ectoderm, mesoderm, and endoderm, forming specialized tissues and organs. While protein-coding genes are established regulators, long non-coding RNAs (lncRNAs) have emerged more recently as potential players in lineage specification. This study establishes an experimental pipeline to examine the role of different lncRNAs—RMST, C13Rik, and 913Rik—in mouse embryonic stem cell (mESC) differentiation, with a focus on ectodermal and mesodermal commitment.

Methods: To explore the functions of these lncRNAs, we optimized 2D differentiation protocols to guide mESCs toward ectodermal and mesodermal lineages. Using antisense oligonucleotides (ASOs), we knocked down either RMST, C13Rik, or 913Rik during 2D differentiation and 3D gastruloid formation. We assessed lineage markers SOX2 (ectoderm) and Brachyury (T) (mesoderm) by using a dual reporter cell line, fluorescence microscopy and fluorescence-activated cell sorting (FACS). Finally, we evaluated gastruloid morphology and lineage marker expression patterns with confocal imaging. **Results:** Our optimized 2D protocols yielded efficient ectodermal and mesodermal differentiation. FACS analysis showed that RMST and C13Rik knockdown reduced the percentage of SOX2-positive cells during ectodermal differentiation and increased the percentage of Brachyury (T)-positive cells. In contrast, 913Rik knockdown decreased Brachyury (T) expression while increasing SOX2 levels. **Conclusion:** RMST and C13Rik appear to play a role in ectodermal differentiation, while 913Rik influences mesodermal commitment. Although preliminary, these findings highlight the potential roles of these lncRNAs in guiding stem cell fate. Future total RNA sequencing studies will elucidate the molecular pathways they regulate, enhancing our knowledge of their role in early development.

Keywords: Long non-coding RNAs (lncRNA); Ectoderm; Mesoderm; Mouse embryonic stem cells (mESCs); Antisense oligonucleotides (ASOs); gastruloids; SOX2; Brachyury (T).

Introduction

Early cell fate commitment during embryonic development is a foundational process that drives the formation of specialized tissues and organs in multicellular organisms¹. In the initial stages of development, pluripotent cells in the embryo begin to differentiate and commit to specific lineages, ultimately forming the three primary germ layers: ectoderm, mesoderm, and endoderm². Each germ layer gives rise to distinct tissues, with the ectoderm developing into the nervous system and sensory organs, the mesoderm forming muscles, bones, and the cardiovascular system, and the endoderm producing organs such as the liver, pancreas, and thyroid³. Disruptions in germ layer specification can lead to severe developmental abnormalities, including spina bifida from ectodermal defects⁴ or congenital septal defects due to mesodermal abnormalities⁵. In some cases, these disruptions can also lead to embryonic lethality⁶, highlighting the importance of understanding the molecular mechanisms governing these processes.

Previous research on the transcriptional networks playing a role during embryo development has primarily focused on protein-coding genes, but recent advances have revealed significant roles for non-coding RNAs, such as long non-coding RNAs (lncRNAs)⁷⁻¹⁰. LncRNAs are defined as RNA molecules over 200 nucleotides long that do not encode proteins¹¹. Once dismissed as "junk" RNA, lncRNAs are now known to play diverse biological functions, such as transcriptional regulation, post-transcriptional control, and epigenetic modification¹³. For example, lncRNAs can direct chromatin-modifying complexes to particular genomic locations¹⁴, influence the stability of mRNAs^{15, 16}, and interact with transcription factors to regulate gene networks¹⁷. Conventional single-cell RNA sequencing techniques primarily target polyadenylated (poly-A) tails of RNAs, a strategy that excludes many lncRNAs^{18, 19}. However, with improved sequencing technologies, such as Smart-seq-total²⁰ and VASA-seq (Vast Transcriptome Analysis of Single Cells by dA-tailing)²¹, the scope for identifying and studying lncRNAs has expanded dramatically. However, despite these technological advancements, many lncRNAs remain functionally uncharacterized.

Studies show that certain lncRNAs are expressed in a stage- and tissue-specific manner during embryonic development²²⁻²⁴, implying distinct regulatory roles during cell fate commitment. For example, the lncRNA Fendrr is involved in mesodermal

lineage differentiation²⁵. It acts as a scaffold for histone-modifying complexes such as PRC2 and TrxG/MLL, which regulate the epigenetic landscape of genes critical for mesoderm development, such as *Foxf1*. *Fendrr* knockdown leads to defects in the mesoderm-derived tissues and impairs proper heart and body wall development²⁵. Another example is *lncR492*, which suppresses the expression of early neural ectoderm markers such as *SOX1* thereby inhibiting neural ectoderm differentiation in mouse embryonic stem cells (mESCs)²⁶.

Our study specifically aims to elucidate the roles of three lncRNAs—RMST, 9130410C08Rik (913Rik), and C130026I21Rik (C13Rik)—in early cell fate commitment and germ layer specification. These lncRNAs were selected based on their expression patterns observed in VASA-seq data from both *in vitro* gastruloid models and in embryos at embryonic day 8.5 (E8.5)²⁷. RMST and C13Rik were found to localize within ectodermal tissues, while 913Rik was localized in mesodermal regions, suggesting their potential involvement in these specific developmental lineages (Supplementary Data, Figure 1). While prior studies have shown that RMST interacts with *SOX2*, a transcription factor essential for pluripotency maintenance and neural lineage specification²⁸, the roles of 913Rik and C13Rik remain largely unexplored. By investigating these lncRNAs, we aim to understand their contributions to cell fate commitment, specifically within ectodermal and mesodermal differentiation.

To explore the functions of these lncRNAs in germ layer formation, we utilized both 2D and 3D culture models of mESCs to examine cell differentiation. We optimized 2D differentiation cultures to enable controlled lineage specification, allowing us to examine ectodermal and mesodermal differentiation independently²⁹. On the other hand, 3D gastruloid models serve as a powerful model to study cell fate decisions and patterning events that recapitulate key events of gastrulation, the critical stage in embryonic development where the germ layers are established³⁰.

Our approach combines these differentiation models with antisense oligonucleotides (ASOs)³¹ to knock down RMST, 913Rik, and C13Rik during mESC differentiation and gastruloid formation. Our initial findings indicate that RMST and C13Rik contribute to ectodermal development, while 913Rik is involved in mesodermal differentiation, in line with their corresponding observed expression patterns in both

embryos and gastruloids. These results mark a preliminary step toward unravelling *in vitro* the molecular mechanisms these lncRNAs regulate germ layer development.

Materials and Methods

Cell culture

Mouse embryonic stem cells (mESCs) genetically engineered to express Sox2::H2B-Venus and T (Brachyury)::H2B-mCherry fluorescent reporters (referred to as Sox2/T mESCs) were kindly provided by the Bernhard Herrmann laboratory at the Max Planck Institute.

Sox2/T mESCs were cultured on plates coated with 0.1% gelatin (Sigma-Aldrich, #G1393) in a humidified incubator at 37°C with 5% CO₂. The cells were cultured in KnockOut™ DMEM medium (Gibco, # 10829018) supplemented with 15% Fetal Bovine Serum (FBS), 1% MEM Non-Essential Amino Acids Solution (100X) (Gibco, #11140050), 1% GlutaMAX, 0.2% ESGRO® Recombinant Mouse LIF Protein (Sigma-Aldrich, #ESG1107), and 0.1% Beta-mercaptoethanol 55mM (Gibco, #31350010).

The culture medium was refreshed daily, and cells were passaged every other day using TrypLE™ Express Enzyme (ThermoFisher Scientific, #12605010) at a 1:6 ratio.

Sox2/T mESCs were regularly tested for mycoplasma contamination. All experiments were conducted with cells with a maximum of 30 passages, with a difference between them within a range of 36-75 passages.

Differentiating mESCs into ectoderm or mesoderm germ layers

Mesoderm differentiation

The mesoderm differentiation protocol was adapted from Winzi M, Casas, 2018²⁶. Sox2/T mESCs were seeded at a density of 15,000 cells/ cm². The differentiation medium consisted of DMEM, low glucose, GlutaMAX (Gibco, #10567014) supplemented with 2% B-27™ Supplement (50X) minus vitamin A (Gibco, #12587010), 1%

NDiff Neuro-2 Medium Supplement (100x) (Gibco, #17502048), 1% MEM Non-Essential Amino Acids Solution (100X) (Gibco, #11140050) and 0.1% Beta-mercaptoethanol 55mM (Gibco, #31350010).

The differentiation protocol took 4 days, with the culture medium replaced with fresh medium containing 3 μ M CHIR-99021(Chiron) (Stem cell technologies, #72054) on days 2 and 3.

Ectoderm differentiation

The ectoderm protocol was modified from Winzi M, Casas, 2018²⁵. Sox2/T mESCs were cultured for six days using the same base medium and seeding density as the mesoderm differentiation protocol, adding 1 μ M retinol to the culture medium. The medium was replaced daily starting from days 2-5 of the differentiation.

Gastruloids formation

Gastruloids were made following the previously established protocol³². In short, mESCs were cultured to 80-90% confluency. 400 cells per well were seeded in 40 μ l of gastruloid medium in a 96-well human U-bottom plate (Greiner, #650160). The gastruloids medium consisted of equal volumes of Neurobasal™ Medium (Gibco, #21103049) and DMEM/F-12 (Gibco, #21331020) supplemented with MEM Non-Essential Amino Acids Solution (100X) (Gibco, #11140050), 1% Sodium Pyruvate 100mM (Gibco, #11360039), 1% GlutaMAX™ Supplement 200mM (Gibco, #35050038), 1% B-27™ Supplement (50X) minus vitamin A (Gibco, #12587010), 0.5% NDiff Neuro-2 Medium Supplement (100x) (Gibco, #17502048), 0.18% Beta-mercaptoethanol 55mM (Gibco, #31350010), and 300 nM Retinol.

On day 2, 150 μ l of fresh medium containing 3 μ M Chiron was added. On days 3 and 4, 150 μ l of the old medium was replaced with fresh medium. The gastruloids were harvested on day 5 for further analysis.

RNA Isolation

RNA was isolated using TRIzol (Invitrogen, #15596026). Briefly, 750 μ l of TRIzol was added to the cell pellet, thoroughly mixed, and incubated at room temperature for 5 minutes. After solubilization, 200 μ l of chloroform was added, followed by centrifugation at 12,000 g for 15 minutes at 4°C leading to phase separation of DNA, protein,

and RNA. The aqueous layer containing RNA was transferred to a new tube. Next, RNA was precipitated using 375 μ l of 100% isopropanol, incubated at room temperature for 10 minutes, and centrifuged for 10 minutes at 4°C. Supernatant was removed. The RNA pellet was washed twice with 1.5 ml of ice-cold 75% ethanol and centrifuged again for 10 minutes at 4°C. The pellet was air-dried at room temperature for 15 minutes. The dried RNA pellet was resuspended in 16 μ l of nuclease-free water. To remove residual DNA, 2 μ l of DNase I (Sigma-Aldrich, #AMPD1) and 2 μ l of DNase buffer (Sigma-Aldrich, #AMPD1) were added and incubated for 20 minutes at 30°C. Next, 2 μ l of DNase stop solution (Sigma-Aldrich, #AMPD1) was added and incubated at 65°C for 10 minutes. Finally, RNA concentration was measured using NanoDrop Spectrophotometer.

cDNA synthesis

500 ng of isolated RNA was mixed with 2 μ l of 50 ng/ μ l random primers (Promega, #C1181), 1 μ l of dNTP mix (ThermoFisher Scientific, #R0191), and nuclease-free water to a final volume of 13 μ l. The mixture was heated at 65°C for 5 minutes and then cooled on ice for 1 minute.

Next, 4 μ l of 5X first-strand buffer, 1 μ l of 0.1 M DTT, 1 μ l of RNase inhibitor, and 1 μ l of SuperScript III reverse transcriptase (all from Invitrogen, #18080093) were added to the reaction mixture. The reaction was incubated at 50°C for 60 minutes, followed by 60°C for 15 minutes in a thermocycler.

RT- qPCR

cDNA was first diluted 1:10. 2 μ l of the diluted cDNA was combined with 5 μ l of SYBR Green qPCR Master Mix (BioRad, #1725279), 1 μ l of 3 μ M of forward and reverse primers (Supplementary Table 1), and 1 μ l of nuclease-free water. The mixture was run on a qPCR machine under the following conditions: 94°C for 10 minutes, 40 cycles of 95°C for 15 seconds, 60°C for 30 seconds, 72°C for 10 seconds, and a final step of 65°C for 5 seconds.

Antisense Oligonucleotides (ASO) treatment

Custom antisense oligonucleotides (ASOs) were designed by Integrated DNA Technology (IDT) to target the lncRNAs RMST, 9130410C08Rik, and C130026I21RIK. Three ASOs per lncRNA were selected based on the location and splice variants of each lncRNAs (Supplementary Table 2). Additionally, a positive control targeting the HPRT housekeeping gene and a negative control targeting a random sequence were included. The ASOs used were in a "gapmer" format, featuring five 2'-O-methyl (2'-OMe) modified bases at each end and ten unmodified DNA bases in the central region.

ASOs were delivered to the cells using Lipofectamine RNAiMAX Transfection Reagent (Invitrogen, #13778100) through reverse transfection at day -1 of the protocol and forward transfections at day 2 for the mesoderm protocol and day 3 for the ectoderm protocol according to the manufacturer's instructions. The final concentration of ASOs used for mesoderm and ectoderm differentiation, and gastruloids formation, was 60 nM.

CellTiter-Glo

To assess cell viability after ASOs treatment, CellTiter-Glo reagent (Promega, #G7570) was diluted 1:4 in PBS. Next, 100 µl of the diluted reagent was added to each well of a 96-well white plate containing SOX2/T mESCs treated with different concentrations of positive control ASOs. After 10-minute incubation at room temperature, luminescence was measured using a GloMax microplate reader with the lid removed.

Imaging

Phase-contrast and (red/green) fluorescence imaging

The IncuCyte S3 Live-Cell Analysis system (Sartorius) was used for phase contrast and red/green fluorescence imaging to monitor mesoderm and ectoderm differentiation over the time course of the differentiation or at different timepoint of differentiation. Sox2::H2B-Venus (green) served as the ectodermal marker, while T (brachyury)::H2B-mCherry (green) was used as the mesodermal marker. Images were captured at 10X magnification and analyzed with IncuCyte Software v2019B.

Confocal microscopy for gastruloids

On day 4 of gastruloids formation, gastruloids were collected and embedded in 10% Matrigel Matrix (Corning, #354234). On day 5, 10 gastruloids were fixed in each well of a 6-well plate using 4% paraformaldehyde and incubated at 4°C overnight. The following day, the gastruloids were stained with Hoechst 33342 (ThermoScientific, #62249) at a concentration of 1:3000 for 24 hours. Before imaging, the gastruloids were transferred into SuperFrost plus slides (Avantor, #631-0108) and mounted with 50 μ l of 80% glycerol. Finally, the gastruloids were imaged with Dragonfly 500 microscope capturing 3D z-stacks.

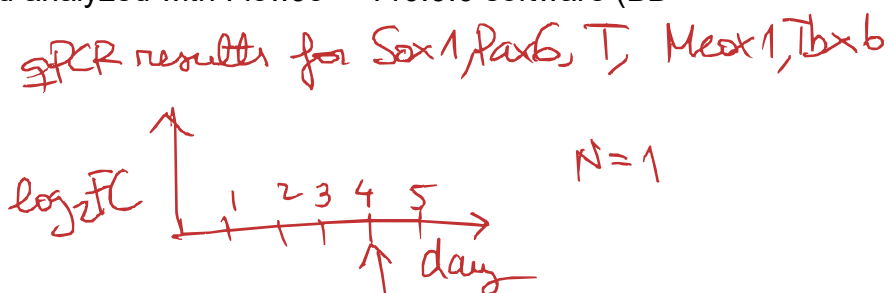
Fluorescence-activated cell sorting (FACS)

At the end point 4 days for Mesoderm, 5 days for Gastruloids, and 6 days for ectoderm. Cells were collected and made into single cell suspensions to FACS sort based on the fluorescence of Sox2::H2B-Venus and T (brachyury)::H2B-mCherry reporters. DAPI was used as a live/dead marker at a concentration of 1:5000. The detailed gating strategy is found in (Supplementary Figure 2).

Unstained undifferentiated mESCs were used as control cells, along with DAPI-negative samples to set the gates for live/dead staining with DAPI. Differentiated mESCs lacking reporter expression were used to set the gate for red (mCherry) and green (Venus) negative fluorescence. Flow cytometry data were acquired using the CytoFlex SRT (Beckman Coulter) and analyzed with FlowJo™ v10.9.0 software (BD Life Sciences).

Results

Ectoderm differentiation protocol optimization: The addition of retinol enhances ectoderm differentiation. To optimize an efficient ectoderm protocol using SOX2/T mESCs, we initially tested a homemade differentiation medium derived from established protocols²⁶. However, this medium exhibited increased expression of the mesodermal markers Brachyury (T), MEOX1, and TBX6 (Supplementary Figure 3). We modified the protocol to address this by supplementing the ectoderm homemade



homemade +

medium with 1 μ M retinol. We then compared the retinol-supplemented medium to a commercially available neural differentiation medium (NDiff 227)³³ over six days to assess their efficacy in promoting ectodermal differentiation.

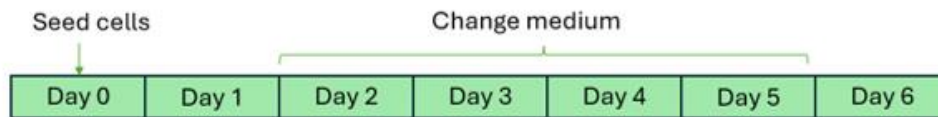
To monitor ectoderm differentiation, we analyzed the expression of key ectoderm markers SOX1 and PAX6 using RT-qPCR (Figure 2b). SOX1 expression gradually increased in both the homemade ectoderm differentiation retinol-supplemented medium and the commercial differentiation medium (NDiff 227), though the increase was more consistent and sustained in the retinol medium. In contrast, PAX6 expression showed a higher and more rapid increase in the retinol medium, especially from day 1 to day 4. The commercial medium, however, displayed a less consistent PAX6 expression pattern, peaking around day 4 and sharply decreasing by day 6.

Next, we analyzed the expression of mesoderm markers Brachyury (T), TBX6, and MEOX1 to assess the suppression of mesodermal differentiation (Figure 2c). The homemade ectoderm differentiation retinol-supplemented and commercial media (NDiff 227) steadily decreased Brachyury (T) expression over time. MEOX1 expression initially dipped in both media; however, expression in the retinol medium remained relatively stable with a slight decrease after the initial drop, while the commercial medium displayed more fluctuation, eventually returning closer to baseline by day 6. Expression of TBX6 remained near baseline across both conditions, with only minimal fluctuations. Notably, the commercial medium exhibited a slight increase in TBX6 starting from day 3, whereas the retinol medium showed slight variability but remained close to zero.

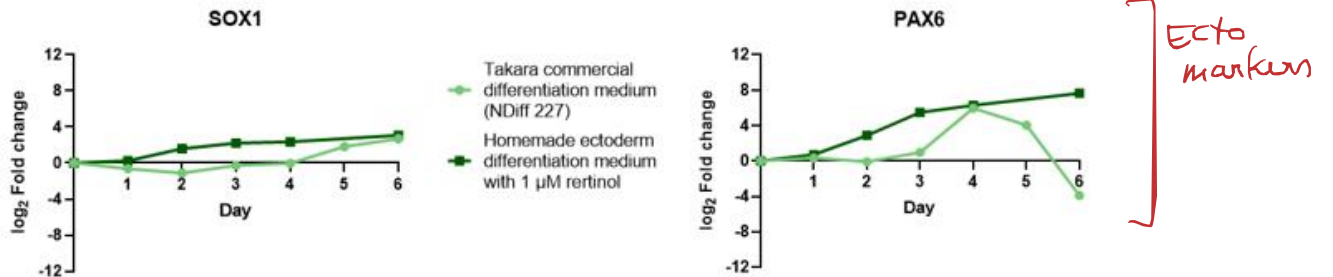
To visually validate these findings, we examined cell morphology and lineage marker expression at day 6 using phase-contrast and fluorescence microscopy (Figure 2d). More cells cultured in the retinol-supplemented homemade ectoderm differentiation medium displayed morphology consistent with ectodermal differentiation than those in the commercial medium (NDiff 227). Fluorescence imaging revealed strong expression of SOX2 (GFP) and minimal Brachyury (T) (mCherry) signal, supporting the RT-qPCR results. In contrast, cells in the commercial medium showed increased Brachyury (T) (mCherry) expression.

→ CONCLUSION: Retinol supplemented is better

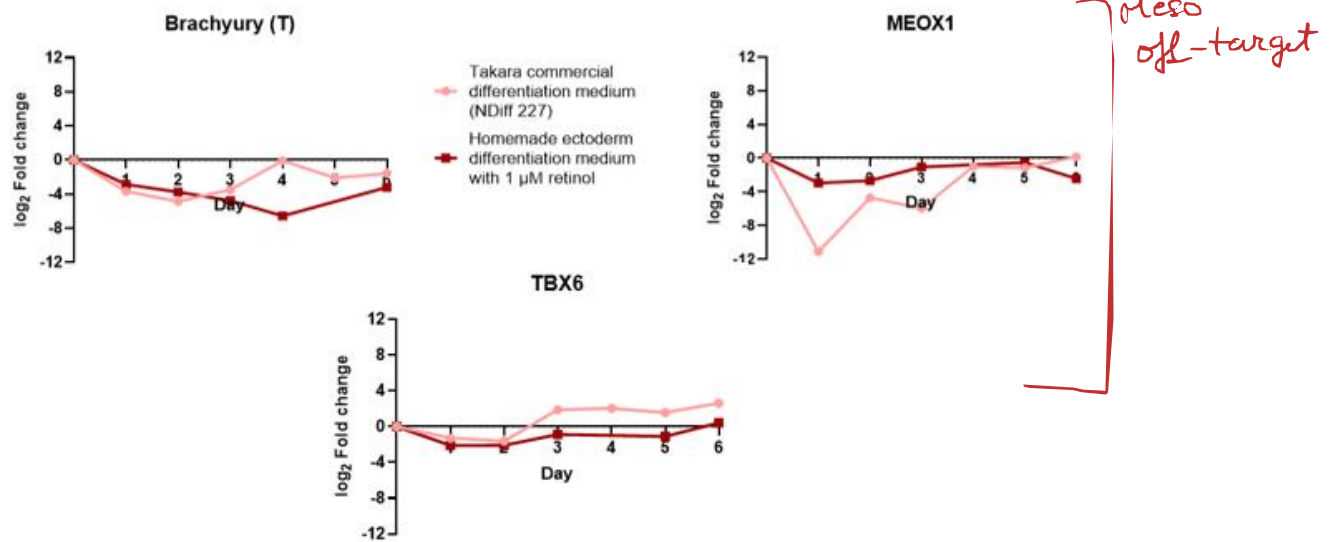
a



b



c



d

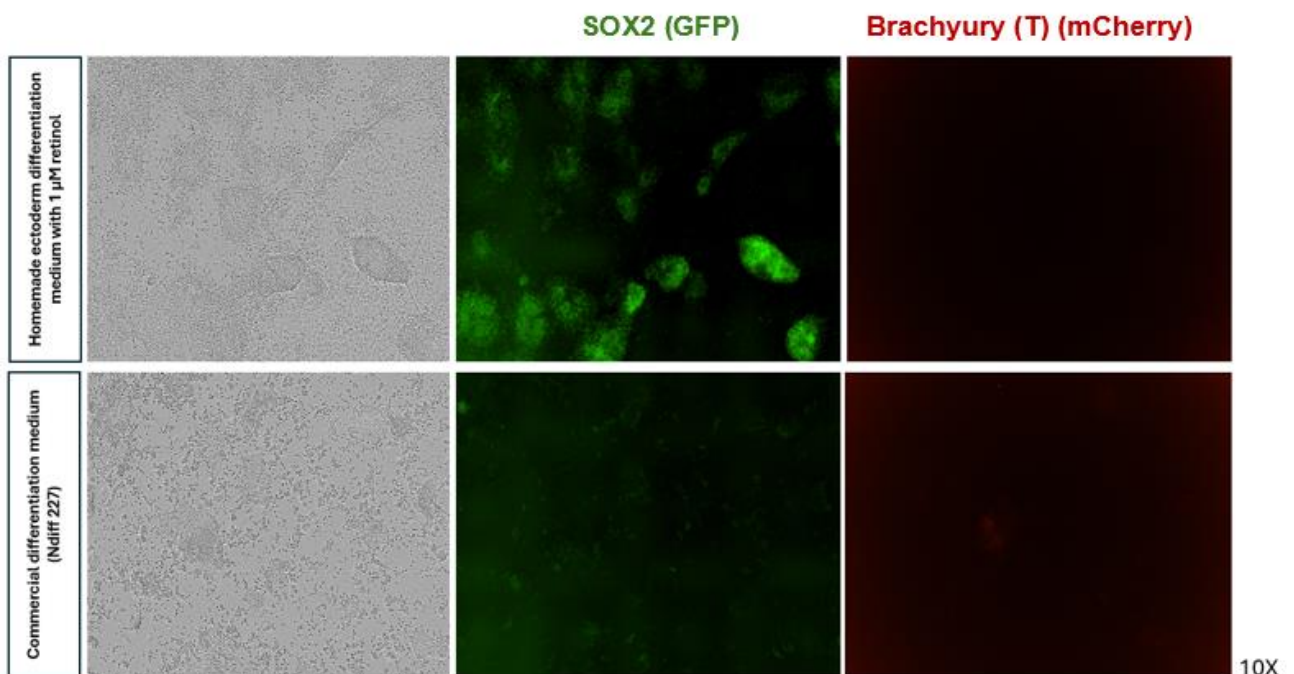


Figure 2 | Ectoderm differentiation protocol optimization. **a** Illustration of the experimental setup and timeline. mESCs were plated at a density of 15,000 cells/ cm² and cultured in ectoderm differentiation medium for 6 days. The medium was changed daily from days 2-5. **b** Time-course RT-qPCR analysis of ectoderm markers (*SOX1*, *PAX6*) and **c** RT-qPCR analysis of mesoderm markers (*Brachyury*, *TBX6*, and *MEOX1*) over 6 days, comparing homemade medium with 1 μM retinol to a commercial neural differentiation medium (Ndiff 227). Log₂ fold change was calculated using the following equation $\text{Log}_2\text{Fold Change} = \log_2(2^{-\Delta Cq_{\text{gapdh}}}/2^{-\Delta Cq_{\text{target}}})$. **d** Representative phase-contrast images (right) showing the morphology of mESCs at day 6 in homemade ectoderm differentiation medium with retinol compared to a commercial ectoderm differentiation medium. Corresponding fluorescence images (left) (Green: *SOX2*; Red: *Brachyury* (T)) displaying expression of lineage markers under the same conditions. Images were captured at 10X magnification using the Incucyte S3 Live-Cell Analysis System. Scale bar: 400 μm. All data represent one experimental repeat (n=1). No statistics were applied.

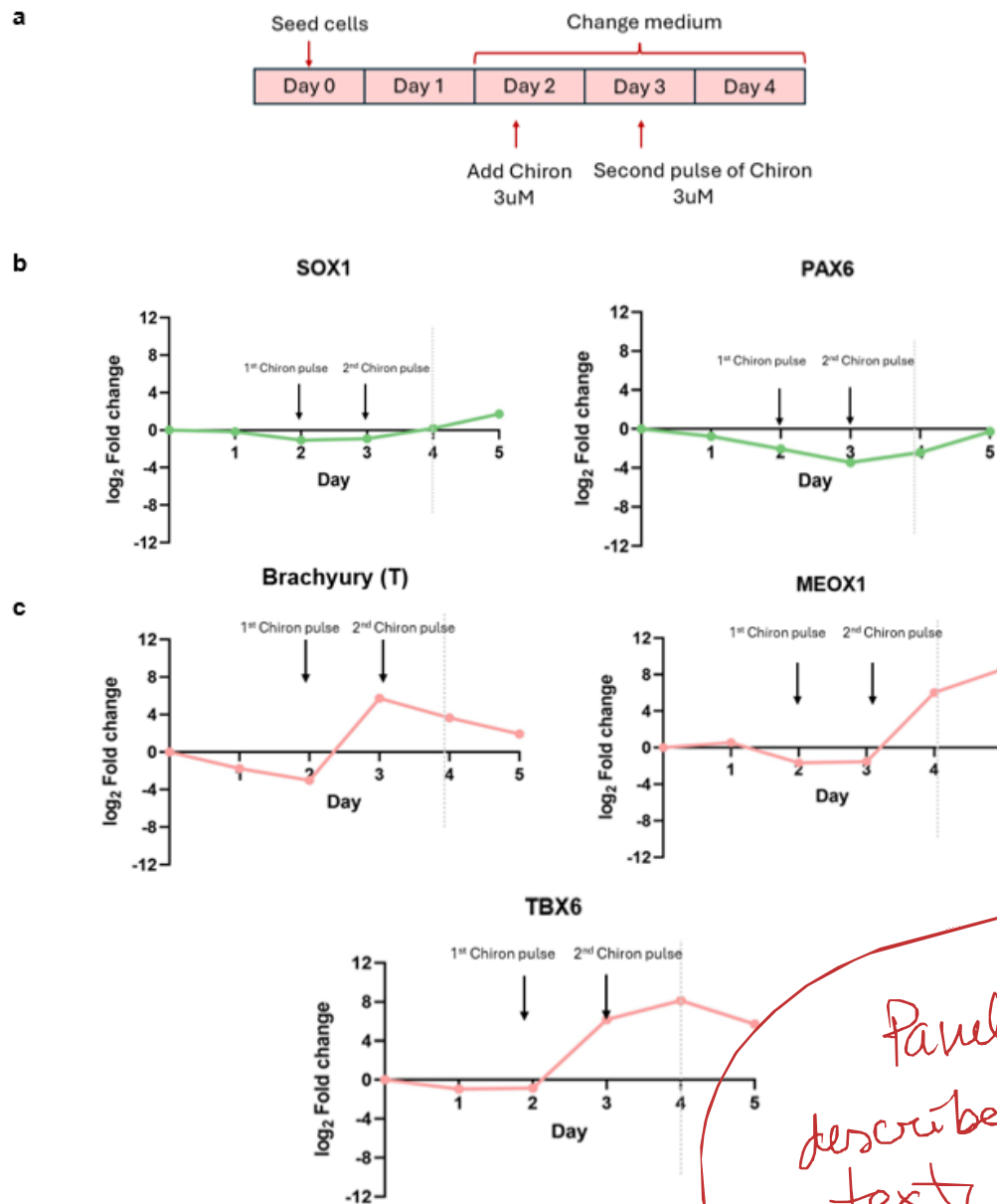
Mesoderm differentiation protocol optimization: More efficient mesoderm differentiation with 2 treatments of 3 μM Chiron. To establish an efficient mesoderm differentiation protocol, we initially tested several published protocols^{26,34-36}. Among the tested methods, treatment with 3 μM Chiron on days 2 and 3 of differentiation produced the most reliable results based on RT-qPCR analysis of mesoderm-specific markers (*Brachyury* (T), *MEOX1*, and *TBX6*). Hence, we chose it for additional optimization (Supplementary Figure 4).

We first monitored the expression of ectoderm markers *SOX1* and *PAX6* over five days (Figure 3b); *SOX1* levels remained close to baseline with only minimal fluctuations throughout the differentiation period, displaying a slight increase after Chiron pulses were stopped. *PAX6* expression showed an initial dip around day 3, followed by a return to near-zero levels by day 5.

Nonetheless, mesoderm markers showed a pronounced response to Chiron treatment (Figure 3c). *Brachyury* (T) expression increased on day 2, peaking significantly on day 3. Similarly, *MEOX1* expression peaked on day 3. *TBX6* showed a more gradual upregulation, with a substantial increase between days 3 and 4.

We observed cell morphology and lineage marker expression at day 4 to visually confirm these findings using phase-contrast and fluorescence microscopy (Figure 3d). Cells cultured with Chiron exhibited morphological changes consistent with mesodermal differentiation. Fluorescence imaging showed robust expression of *Brachyury* (T) (mCherry) and low levels of *SOX2* (GFP), which is in line with the RT-qPCR data.

Log₂FC with different protocols
n=1 again
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Panel d,
described on the
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Figure 3 | Mesoderm differentiation protocol optimization. **a** Illustration of the experimental setup and timeline. mESCs were plated at a density of 15,000 cells/ cm² and cultured in differentiation medium for 4 days. Chiron (3 μM) was added with medium changes on days 2 and 3. **b** Ectoderm markers (SOX1 and PAX6) expression was analyzed using RT- qPCR over 5 days. **c** Mesoderm markers (Brachyury (T), TBX6, and MEOX1) were also analyzed by RT-qPCR under the same conditions. Log₂ fold change was calculated using the following equation $\log_2 \text{Fold Change} = \log_2 \left(\frac{2^{-\Delta Cq_{\text{gapdh}}}}{2^{-\Delta Cq_{\text{target}}}} \right)$. **d** Representative phase-contrast images (right) showing the morphology of mESCs at day 4. Corresponding fluorescence images (left) (Green: SOX2; Red: Brachyury (T)) displaying expression of lineage markers under the same conditions. Images were captured at 10X magnification using the Incucyte S3 Live-Cell Analysis System. Scale bar: 400 μm. All data represent one experimental repeat (n = 1). No statistics were applied.

Optimization of ASOs delivery: Enhanced transfection efficiency and reduced cytotoxicity using Lipofectamine 3000. To optimize antisense oligonucleotide (ASO) treatment conditions, we used the positive control ASO targeting the HPRT gene to compare efficacy across different transfection techniques (Figure 4b). ASOs were delivered to mESCs at three concentrations (0.5, 5, and 20 nM) using either a naked transfection approach, Lipofectamine RNAiMAX, or Lipofectamine 3000.

To evaluate the cytotoxic effects of each transfection method and ASO concentration, we evaluated mESCs cell viability 24 hours post-treatment with the positive control ASOs. Results showed a dose-dependent decrease in viability across all concentrations and methods (Figure 4c). However, Lipofectamine 3000 and RNAiMAX transfections preserved higher cell viability at all ASO concentrations than naked ASO delivery.

To assess the duration of the knockdown effect of antisense oligonucleotides (ASOs), mESCs were treated with 60 nM of a positive control ASO targeting HPRT. Ectoderm differentiation was initiated, and cells were collected daily for four days for RT-qPCR analysis of HPRT expression. The results, represented as log₂ fold changes, indicate a sharp reduction in HPRT levels on Day 0 of differentiation (24 hours after treatment), with a transient knockdown effect observed for 5 days post-treatment (Figure 4d).

this is viability

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→ This is HPRT expression

which is not tested before

lots of fluctuation?
How do you explain this?

What's then the choice for the upcoming experiments? Naked for simplicity?

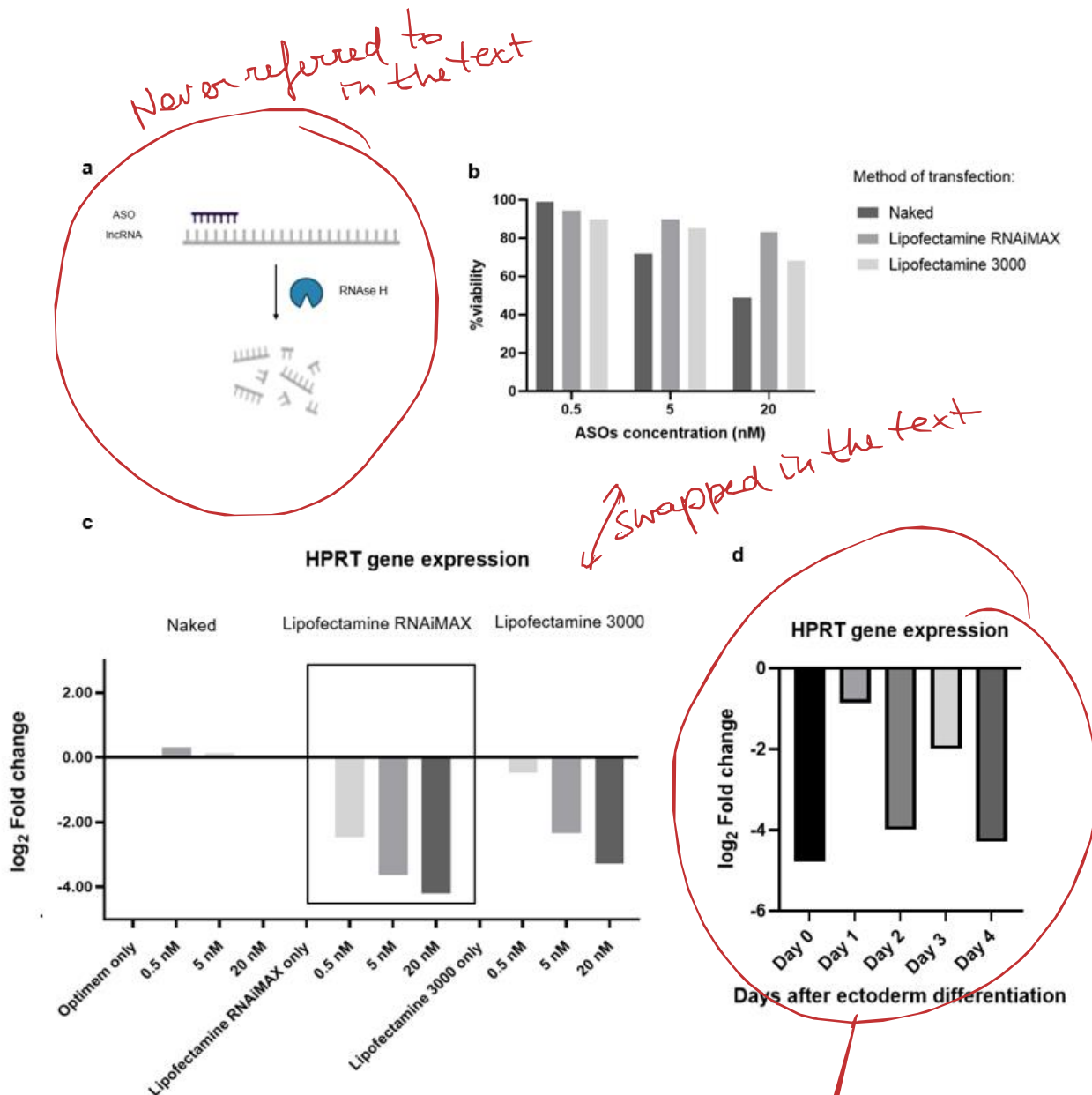


Figure 4 | ASOs treatment optimization. **a** Diagrammatic representation of the mechanism of action of ASOs created in Biorender. **b** mESCs were treated with (0.5, 5, or 20 nM) targeting the housekeeping gene HPRT referred to as positive control ASOs, using different transfection methods: naked transfection, lipofectamine 3000, or Lipofectamine RNAiMAX. Cell viability was measured after 24-hours of treatment using CellTiter-Glo Luminescent Cell Viability Assay. **c** RT-qPCR was performed to analyze HPRT expression under the same treatment conditions. **d** mESCs were treated with 60 μ M ASOs targeting HPRT, and ectoderm differentiation was initiated. Cells were collected daily for four days, and RT-qPCR was performed to measure HPRT expression. The results are shown as log₂ fold changes. Log₂ fold change was calculated using the following equation $\text{Log}_2\text{Fold Change} = \log_2(2^{-\Delta\text{Cq}_{\text{gapdh}}} / 2^{-\Delta\text{Cq}_{\text{target}}})$. All data represent one experimental repeat (n=1). No statistics were applied.

What are the different colors? How do you read the fluctuation seen?

Knocking down RMST and C13Rik during ectoderm differentiation potentially leads to less SOX2 expression and increased Brachyury (T) expression. To investigate the effect of ASOs knockdown of RMST and C13Rik during ectoderm differentiation, mESCs were treated with ASOs 24 hours before differentiation and again on day 3 of differentiation, to address the transient nature of ASO-mediated knockdown. On day 6, we used flow cytometry to assess the expression of SOX2 (GFP) and Brachyury (T) (mCherry) in both treated and control groups. Cells were categorized into four populations based on SOX2 and Brachyury expression: SOX2-positive (ectodermal lineage), Brachyury-positive (mesodermal lineage), SOX2/Brachyury double-positive, and double-negative (Figure 5b).

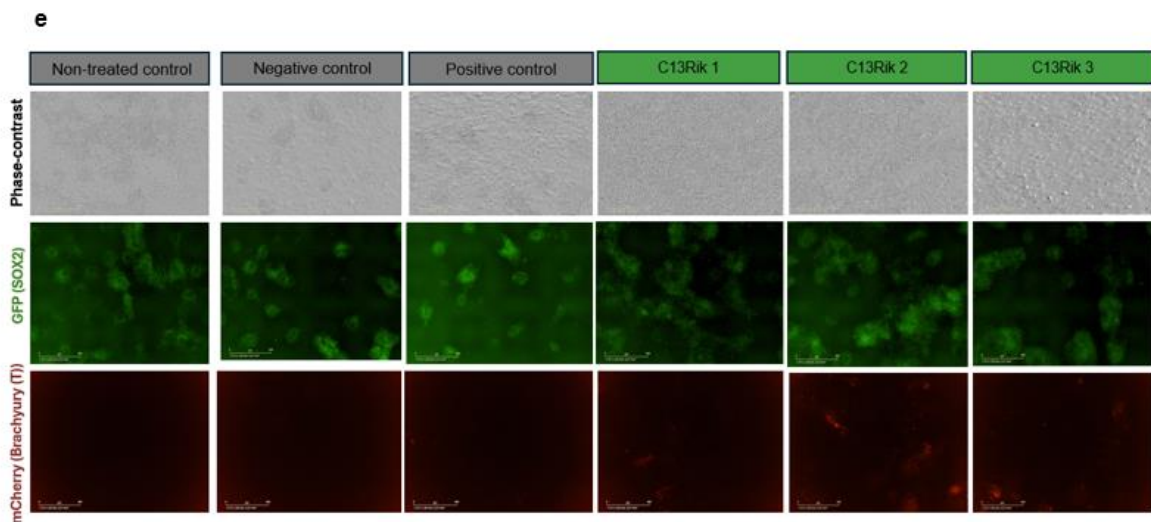
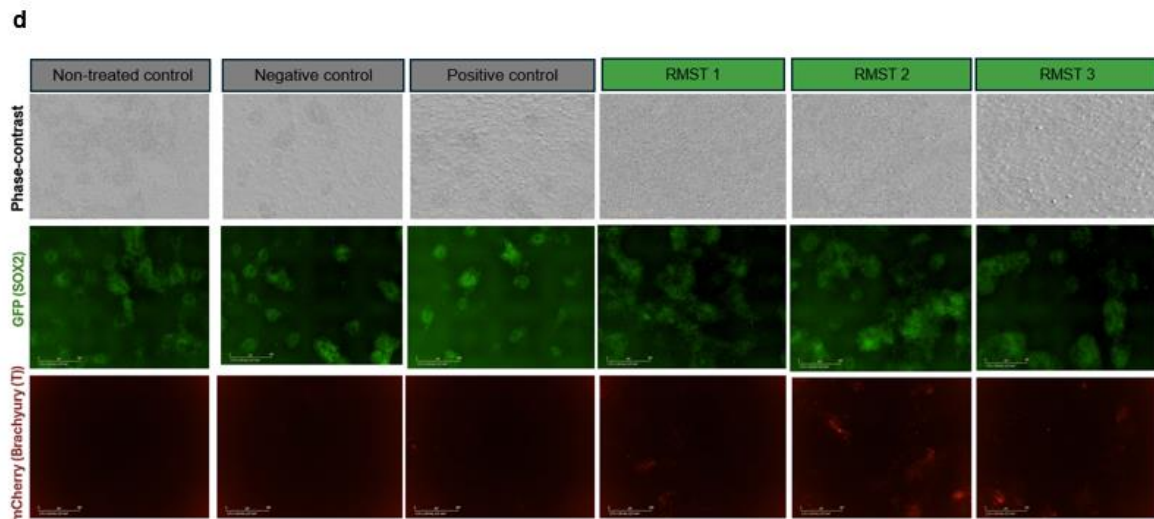
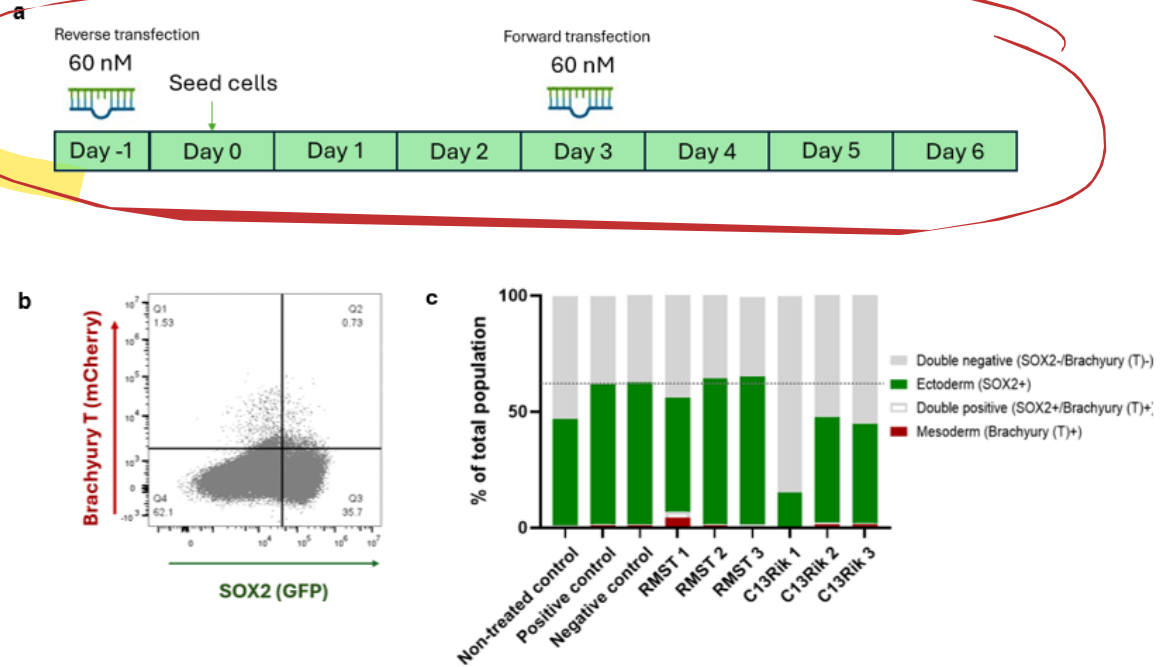
We tested three ASOs designed for each target. Treatment with ASOs targeting RMST showed variable effects, RMST ASO 1 reduced SOX2-positive cells and increased Brachyury-positive populations. Similarly, all three ASOs targeting C13Rik (C13Rik 1, 2, and 3) decreased the proportion of SOX2-positive cells, with C13Rik 1 producing the strongest effects (Figure 5c).

Microscopy images further validated these findings. Phase-contrast images revealed that untreated and control-treated cells maintained typical ectodermal morphology, while cells treated with ASOs targeting RMST (RMST 1) and C13Rik (C13Rik 2 and 3) displayed altered morphology consistent with mesodermal differentiation. Cells treated with RMST 1 and C13Rik 2 and 3 ASOs showed reduced SOX2 expression and increased Brachyury (mCherry) signal (Figure 5d+e).

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never referred to in the text



Scale bar

10X

is this the naked approach? what are the differences

Figure 5 | ASOs treatment targeting RMST and C13Rik during ectoderm differentiation. **a** Illustration of the experimental setup and timeline. mESCs were treated with 60 nM of ASOs via reverse transfection. After 24 hours, cells were counted and plated to begin ectoderm differentiation. On day 3 of differentiation, a second treatment with 60 nM ASOs was administered via forward transfection, and differentiation continued until day 6. **b** Representative scatter plot of the FACS strategy used to distinguish cell populations based on SOX2 and Brachyury (T) expression. **c** Stacked bar graph showing the distribution of SOX2/Brachyury (T) double-negative, SOX2-positive, SOX2/Brachyury (T) double-positive, and Brachyury (T)-positive cells under different treatment conditions, including non-treated control, positive control targeting HPRT, negative control, and three ASOs targeting RMST or C13Rik. **d** Phase-contrast (top) and corresponding fluorescence microscopy images (bottom) illustrating SOX2 (GFP) and Brachyury (T) (mCherry) expression patterns comparing the non-treated control, positive control targeting HPRT, negative control, and three different ASOs targeting RMST on day 6 of differentiation. **e** Phase-contrast and fluorescence images showing three ASOs targeting C13Rik. Images were captured at 10X magnification using the Incucyte S3 Live-Cell Analysis System. Scale bar: 400 μ m. All data represent one experimental repeat (n=1). No statistics were applied.

how are the thresholds decided? what is the dashed horizontal line?

Knocking down 913Rik during mesoderm differentiation appears to reduce Brachyury (T) expression while increasing SOX2 expression. To investigate the effect of ASOs knockdown of 913Rik during mesoderm differentiation, mESCs were treated with ASOs 24 hours prior to differentiation and again on day 2. On day 4, we assessed lineage-specific markers: SOX2 (ectoderm) and Brachyury (T) (mesoderm) using flow cytometry. Cells were categorized into four populations: SOX2-positive, Brachyury-positive, double-positive, and double-negative (Figure 6b).

Among the three ASOs designed and tested for 913Rik, **ASO 3** reduced the Brachyury-positive population compared to the negative control, while increasing the proportion of SOX2-positive cells (Figure 6c).

Fluorescence microscopy supported these results. Cells treated with 913Rik 3 ASOs showed reduced Brachyury signal and increased SOX2 expression (Figure 6d).

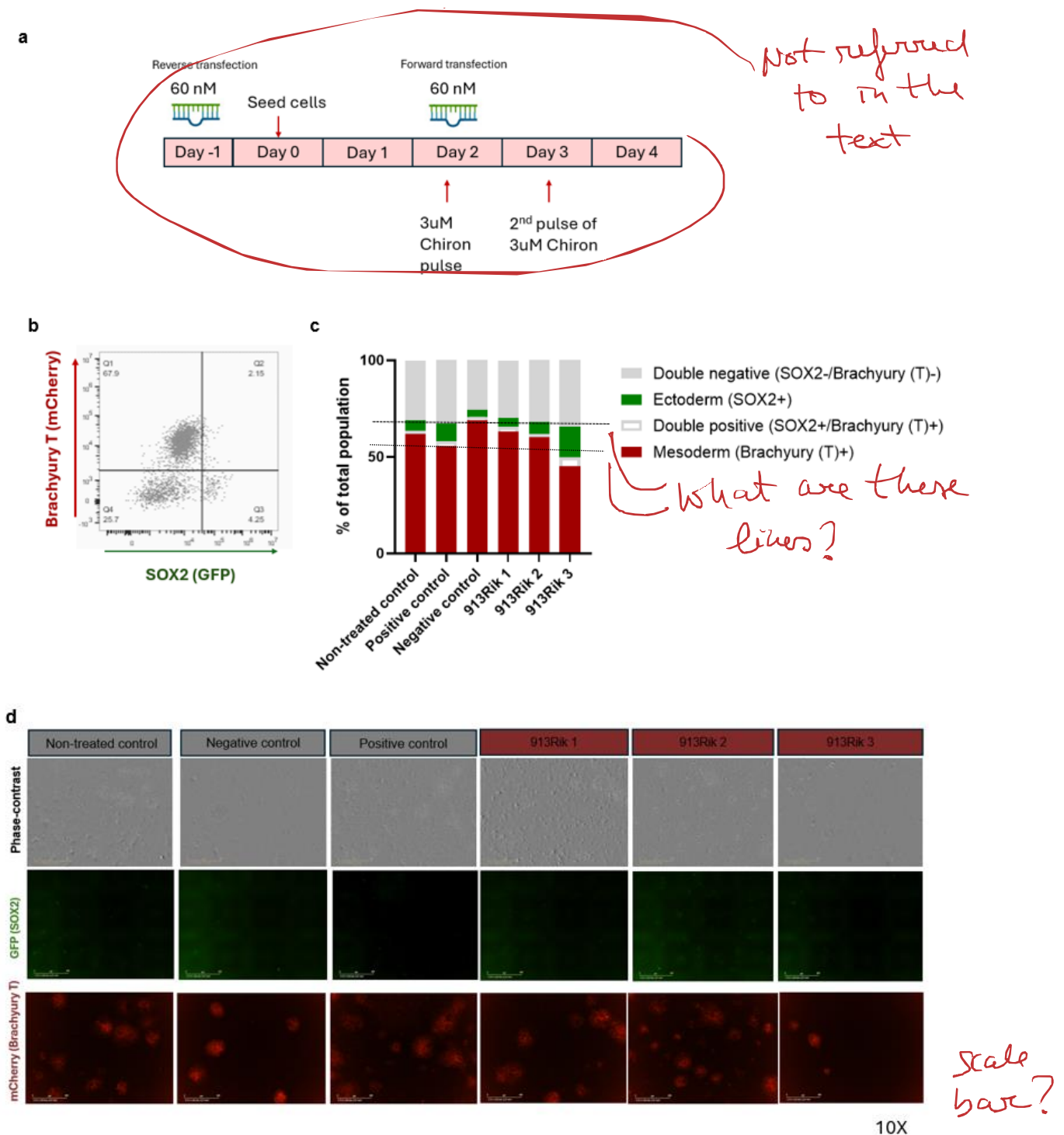


Figure 6 | ASOs treatment targeting 913Rik during mesoderm differentiation. **a** Schematic overview of the experimental setup and timeline. mESCs were treated with 60 nM of ASOs via reverse transfection. After 24 hours, mesoderm differentiation was started. On day 2 of differentiation, a second treatment with 60 nM ASOs was administered via forward transfection. 3 μ M of Chiron were added on days 2 and 3 and differentiation continued until day 4. **b** Representative scatter plot of the FACS strategy used to distinguish cell populations based on SOX2 and Brachyury (T) expression. **c** Stacked bar graph showing the percentage of SOX2/Brachyury (T) double-negative, SOX2-positive, SOX2/Brachyury (T) double-positive, and Brachyury (T)-positive cells under different treatment conditions, including non-treated control, positive control targeting HPRT, negative control, and three ASOs targeting RMST or C13Rik. **d** Phase-contrast (top) and

corresponding fluorescence microscopy images (bottom) illustrating SOX2 (GFP) and *Brachyury* (T) (mCherry) expression patterns comparing the non-treated control, positive control targeting *HPRT*, negative control, and three different ASOs targeting 913Rik on day 4 of differentiation. Images were captured at 10X magnification using the Incucyte S3 Live-Cell Analysis System. Scale bar: 400 μ m. All data represent one experimental repeat (n=1). No statistics were applied.

(the mesoderm one)

Evaluation of 913Rik function in gastruloid formation. To investigate the role of 913Rik in gastruloid development, we initiated gastruloid formation using 3 μ M Chiron, with or without ASOs targeting 913Rik. On day 4, gastruloids were embedded in Matrigel for structural support, and morphological assessment and confocal imaging were performed on day 5 (Figure 7b). ?

Phase-contrast imaging on day 5 revealed that gastruloids treated with the positive control ASO (targeting *HPRT*) and those treated with 913Rik-targeting ASO 1 exhibited smaller morphologies (Figure 7c).

For confocal microscopy, while the number of imaged gastruloids was insufficient to reach definitive conclusions, those treated with 913Rik ASO 3 showed increased SOX2 expression compared to the negative control-treated gastruloids (Figure 7d).

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with previous 2D data

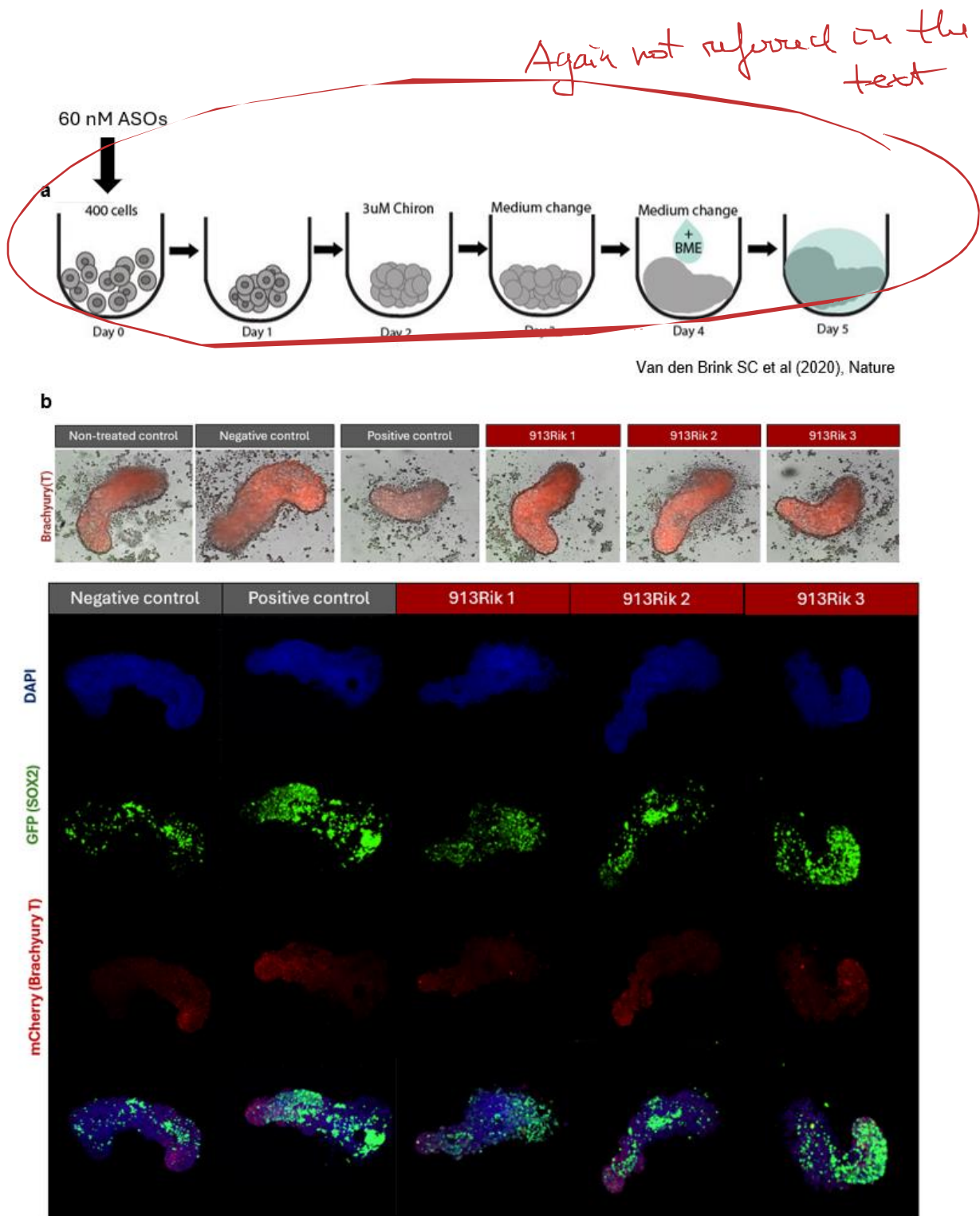


Figure 7 | 913Rik targeting ASOs treatment during gastruloids formation. **a** Experimental setup of Gastruloid formation figure adopted from Van den Brink SC et al. (2020). A total of 400 mESCs were treated with 60 nM of each of the positive control ASOs targeting HPRT, the negative control, and three different ASOs targeting 913Rik and seeded on day 0. On day 2, 3 μ M Chiron was added, and gastruloids were embedded in Matrigel on day 4 for confocal imaging. **b** Fluorescence images illustrating gastruloid morphology and Brachyury (T) expression at day 5 following treatment with different 913Rik-targeting ASOs. Images were captured at 10X magnification using the Incucyte S3 Live-Cell Analysis System. Scale bar: 400 μ m. **c** Confocal images taken under the same conditions as panel b, captured at 10X magnification with the Dragonfly 500. Scale bar: 200 μ m.

where?

Discussion

Retinol enhances ectoderm differentiation protocol

Our findings demonstrate that supplementing an ectoderm homemade differentiation medium derived from published literature²⁵ with retinol (1 μ M) enhances ectodermal differentiation while effectively suppressing mesodermal differentiation. This is supported by RT-qPCR, SOX2 and Brachyury (T) reporter activity and morphological observations and comparisons to a commercially available neural differentiation medium.

The enhanced differentiation observed in the retinol-supplemented medium is likely due to retinol's influence on retinoic acid (RA) signaling pathways. Retinol is metabolized into retinoic acid, which plays a critical role in the regulation of neural lineage through nuclear RA receptors (RARs)³⁷. These receptors regulate the expression of neural genes, including SOX1 and PAX6, driving neural commitment while repressing non-neural fates^{38,39}. In line with this mechanism, the retinol-supplemented medium increased the expression of SOX1 and PAX6, as confirmed by RT-qPCR. SOX1 levels showed a consistent and sustained increase, indicating stability in supporting ectodermal pathways over six days.

Additionally, PAX6 expression exhibited a more rapid and pronounced increase than the commercial medium, particularly during the early differentiation phase (days 1–4). PAX6 is essential for neural tube patterning and neural progenitor specification, suggesting that retinol may accelerate the early commitment of pluripotent stem cells to ectodermal fates⁴⁰.

The retinol-supplemented medium also effectively suppressed mesodermal marker expression, including Brachyury (T), TBX6, and MEOX1, which are critical for mesoderm specification and differentiation. This suppression may reflect the antagonistic relationship between RA signaling and Wnt/ β -catenin pathways, as RA has been shown to inhibit canonical Wnt pathway⁴¹, which is essential for mesodermal lineage commitment⁴².

Microscopic observations further validated these findings. Cells cultured with retinol supplementation exhibited improved ectodermal differentiation morphology.

Fluorescence imaging revealed strong SOX2 (GFP) expression and minimal Brachyury (mCherry) signal which confirms the suppression of mesodermal differentiation. In contrast, the commercial medium demonstrated less robust ectodermal differentiation and higher mesodermal marker expression along with a less strong ectodermal morphological phenotype.

These results collectively demonstrate that retinol supplementation enhances ectodermal differentiation. In contrast, the commercial medium, while effective to some extent, exhibited inconsistencies in marker expression and less reliable suppression of mesodermal differentiation.

Further studies should explore optimizing retinol concentration as the effects of retinoic acid are highly dose-dependent and excessive RA could potentially disrupt normal patterning processes⁴³.

Dual-pulse Chiron treatment improves mesoderm differentiation protocol

To optimize mesoderm differentiation, we first tested established methods, including BMP4, a single pulse of Chiron, and Activin A treatments. While BMP4 and Activin A are widely used for mesodermal differentiation, they can also lead to off-target cell differentiation toward the endoderm lineage⁴⁴. A single Chiron pulse was insufficient to sustain mesoderm differentiation for the full differentiation time scale (4-5 days) RT-qPCR analysis revealed a downregulation in mesodermal markers and an increase in ectodermal markers upon stopping of Chiron treatment which warranted repeated stimulation.

Building on these observations, we developed a protocol using two pulses of 3 μ M Chiron on days 2 and 3. This approach effectively induced mesoderm-specific markers while suppressing ectodermal markers. Chiron, a GSK-3 inhibitor and Wnt activator, activates the Wnt/ β -catenin pathway, a key driver of mesoderm specification³⁶. The dual-pulse protocol upregulated Brachyury (T), MEOX1, and TBX6, with distinct expression dynamics. Brachyury expression began rising on day 2, peaking sharply on day 3, consistent with its role as an early mesodermal marker critical for specification and migration during gastrulation⁴⁵.

Similarly, MEOX1 expression peaked on day 3, reflecting its function in somitogenesis and mesodermal tissue patterning⁴⁶. Unlike Brachyury (T) and MEOX1,

TBX6 showed a more gradual increase, with upregulation between days 3 and 4. TBX6, a transcription factor downstream of Brachyury (T) is crucial for paraxial mesoderm development, but not necessary for the initial lineage decision. This temporal delay relative to Brachyury aligns with its role downstream in the mesoderm differentiation cascade⁴⁷.

Morphological changes and fluorescence imaging of the reporter cell line confirmed mesodermal commitment with strong Brachyury (T) (mCherry) and minimal SOX2 (GFP) signals, supporting RT-qPCR results.

Using two pulses of 3 μ M Chiron, this optimized protocol effectively induces mesoderm differentiation. The coordinated temporal expression of Brachyury, MEOX1, and TBX6 underscores the protocol's precision in recapitulating mesoderm development.

→ but is it the one you used?

Lipofectamine RNAiMAX is the most efficient ASOs delivery method in mESCs

In this study, we used ASOs incorporating five 2'-O-methyl (2'-OMe) bases at each end to enhance stability and improve target binding⁴⁸.

ASOs delivery method is essential for achieving effective gene knockdown while minimizing cytotoxicity, especially in sensitive cell types like mouse embryonic stem cells (mESCs). In this study, we compared three delivery techniques—naked transfection (without any transfecting agent), Lipofectamine RNAiMAX, and Lipofectamine 3000—across various ASO concentrations to assess their transfection efficiency and cytotoxicity. Our findings highlight Lipofectamine RNAiMAX as the most effective option, combining high transfection efficiency with low cytotoxicity, whereas naked transfection proved the least efficient and most toxic method.

Naked transfection, which involves applying ASOs directly without a delivery vehicle, consistently yielded the lowest cell viability across all tested ASO concentrations. This poor performance is likely due to the degradation of naked ASOs by extracellular nucleases which makes them less bioavailable⁴⁹. Furthermore, degraded nucleic acid fragments may accumulate and activate inflammatory and apoptotic pathways which leads to increased cytotoxicity⁵⁰.

While similarly effective in delivering ASOs compared to RNAiMAX, Lipofectamine 3000 showed slightly reduced cell viability at higher concentrations. Although the

manufacturer does not provide specific details on the differences between these lipid reagents, these variations could be due to differences in lipid composition, or the pathways used for the efficient transport of ASOs into the cytoplasm.

Our results demonstrate that Lipofectamine RNAiMAX is the optimal choice for ASO delivery in mESCs, balancing high transfection efficiency with minimal cytotoxic effects.

Knockdown of RMST, C13Rik, and 913Rik demonstrates potential roles in lineage specification during ectoderm and mesoderm differentiation

The knockdown of RMST, C13Rik, and 913Rik using antisense oligonucleotides (ASOs) during ectoderm and mesoderm differentiation led to visible effects, including cell death and alterations in fluorescent reporter activity, as observed in FACS data and reporter activity observations. In the case of RMST and C13Rik, their knockdown resulted in reduced SOX2 (ectodermal marker) expression and increased Brachyury (T) (mesodermal marker) expression, suggesting that these lncRNAs help sustain ectodermal identity and that their downregulation biases differentiation towards mesodermal fates.

The reduction of SOX2-positive cells following RMST knockdown aligns with prior studies that establish RMST as a critical long noncoding RNA involved in SOX2 regulation. RMST interacts with the SOX2 transcription factor which enhances its binding to target genes during neural differentiation⁵¹. Disrupting this interaction with RMST ASOs might impair SOX2's ability to maintain ectodermal lineage commitment, leading to a shift towards mesodermal differentiation. The observed increase in Brachyury (T) expression further supports this interpretation, as SOX2 functions as a gatekeeper of ectodermal lineage specification by antagonizing mesodermal gene programs^{46, 52, 53}.

Similarly, the knockdown of C13Rik reduced SOX2-positive populations and increased Brachyury (T) expression, suggesting its involvement in ectodermal lineage commitment. While the exact mechanisms by which C13Rik influences lineage specification are unclear, these findings suggest its regulatory role in ectodermal differentiation and warrant further investigation.

where do I see statistics

The knockdown of 913Rik during mesoderm differentiation and gastruloid formation revealed its potential role in promoting mesodermal differentiation. Targeting 913Rik with ASOs led to reduced Brachyury (T) expression and increased SOX2 expression. These results suggest that 913Rik positively regulates mesodermal differentiation.

In gastruloids experiments, the functional relevance of 913Rik was further underscored. Gastruloids treated with 913Rik ASO 1 exhibited smaller morphologies, suggesting impaired growth. Confocal imaging results showing increased SOX2 expression in 913Rik ASO 3-treated gastruloids supported the molecular data from the 2D mesoderm differentiation experiment, where SOX2 upregulation coincided with Brachyury suppression.

Limitations of the Study and Outlook

This study provides valuable insights into the potential roles of RMST, C13Rik, and 913Rik lncRNAs in germ layer specification. However, several limitations should be noted. One significant limitation is the narrow scope of this study, focusing on only three lncRNAs. Moreover, we conducted the experiments with only one biological replicate. Larger sample sizes in future studies would increase the reliability of the results and allow for more definitive conclusions.

Another limitation is the variability in ASOs efficacy. Although three ASOs were designed for each target gene, variability in their effects was observed, with some ASOs, like RMST 1 and C13Rik 1, showing stronger phenotypic and molecular outcomes. This may be due to differences in ASO efficiency, off-target interactions, or incomplete knockdown. One reason for ASOs incomplete knockdown or off-target interactions, is that they rely on endogenous RNA degradation pathways such as RNase H⁵⁴. Off-target interactions are particularly concerning in our study as long noncoding RNAs, such as RMST, C13Rik, and 913Rik, have uncharacterized roles and interacting partners.

Additionally, the transient nature of ASO knockdown presents a limitation⁵⁵. Since ASOs provide only a temporary reduction in target gene expression, they may not fully replicate the effects of long-term gene silencing or genetic deletion, potentially

missing downstream compensatory mechanisms. Stable genetic models like knock-out cell lines could be employed to overcome this.

Despite these challenges, ASOs offer a significant advantage over RNA interference for studying lncRNA function. Unlike RNAi, which predominantly targets cytoplasmic RNAs, ASOs can efficiently target both nuclear and cytoplasmic⁵⁶. This is particularly important for our study, as the cellular localization of C13Rik, and 913Rik is unknown.

Using *in vitro* models, including 2D differentiation and gastruloid formation, introduces another limitation. While these systems mimic early embryonic development, they do not fully replicate the complexity of *in vivo* development. Gastruloids more closely resemble key events like mesodermal and ectodermal lineage specification and can capture spatial and temporal patterning⁵⁷. However, they lack extra-embryonic tissues and vascularization present in *in vivo* models⁵⁸. Future studies should incorporate *in vivo* models to validate the physiological relevance of genes like RMST, C13Rik, and 913Rik in lineage specification.

Moreover, SOX2 was used as a marker for ectoderm differentiation in fluorescence observations, but it is also a marker of pluripotent, undifferentiated cells⁵⁹, complicating the interpretation of results. SOX2 expression may reflect both ectodermal differentiation and residual pluripotency.

Finally, while this study characterized phenotypic effects and changes in SOX2 and Brachyury (T) expression following knockdown of RMST, C13Rik, and 913Rik, it did not investigate underlying molecular mechanisms. Detecting these lncRNAs via RT-qPCR was challenging due to their low expression levels⁶⁰ and multiple splice variants. To overcome this, future studies could adopt methods like VASA single-cell sequencing to validate lncRNA expression and better understand their roles in lineage specification.

Outlook

Future research should use techniques like RNA immunoprecipitation (RIP) and chromatin isolation by RNA purification (ChIRP) to uncover interactions between RMST, C13Rik, and 913Rik and proteins, chromatin, or other RNAs. Moreover, it could investigate the potential involvement of these lncRNA in developmental

disorders. *In vivo* models like zebrafish or mice, along with iPSC-derived models of human diseases, could be used to assess the impact of gene knockdown during early development, which could lead to finding therapeutic targets for developmental abnormalities.

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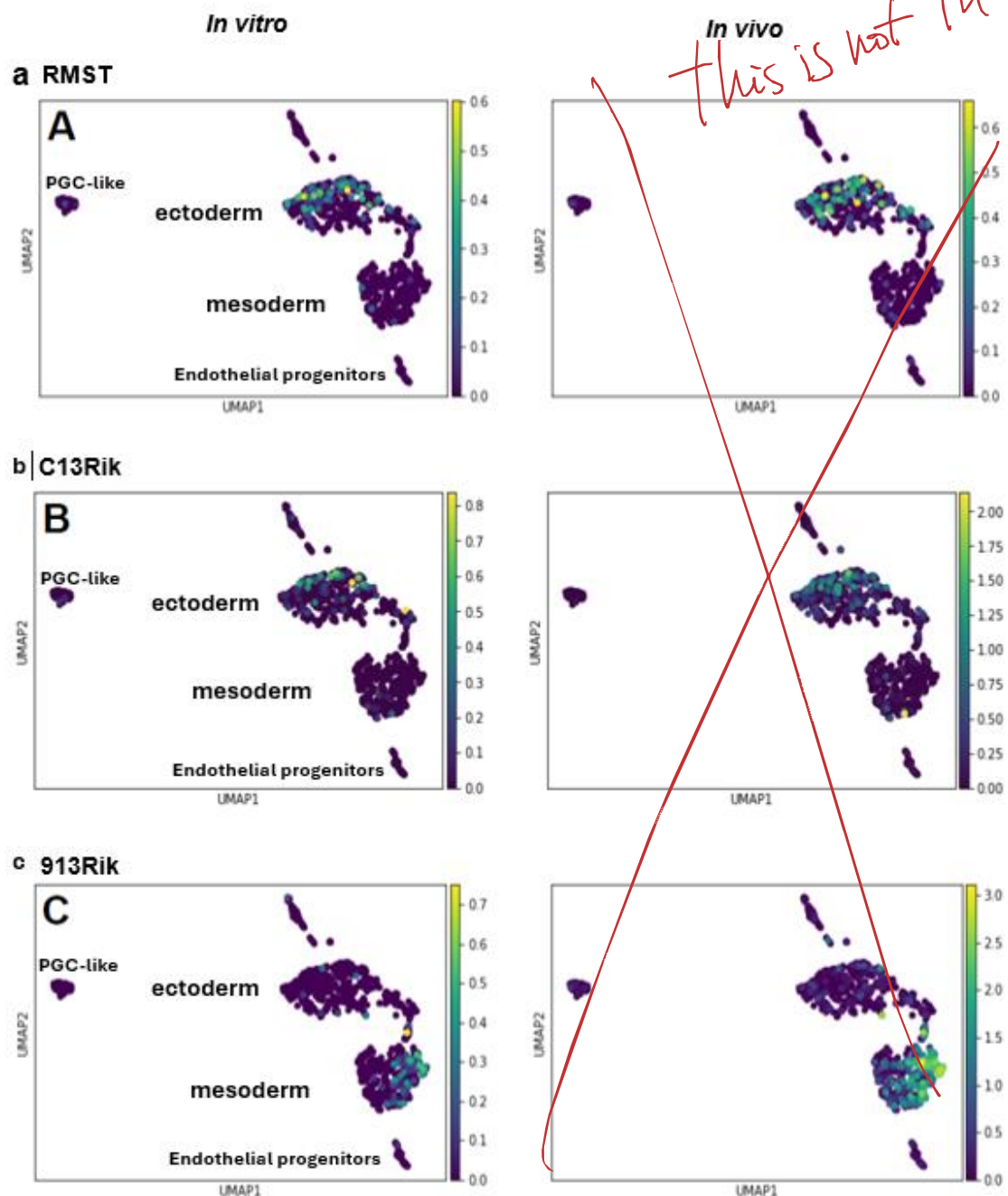
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Supplementary Data



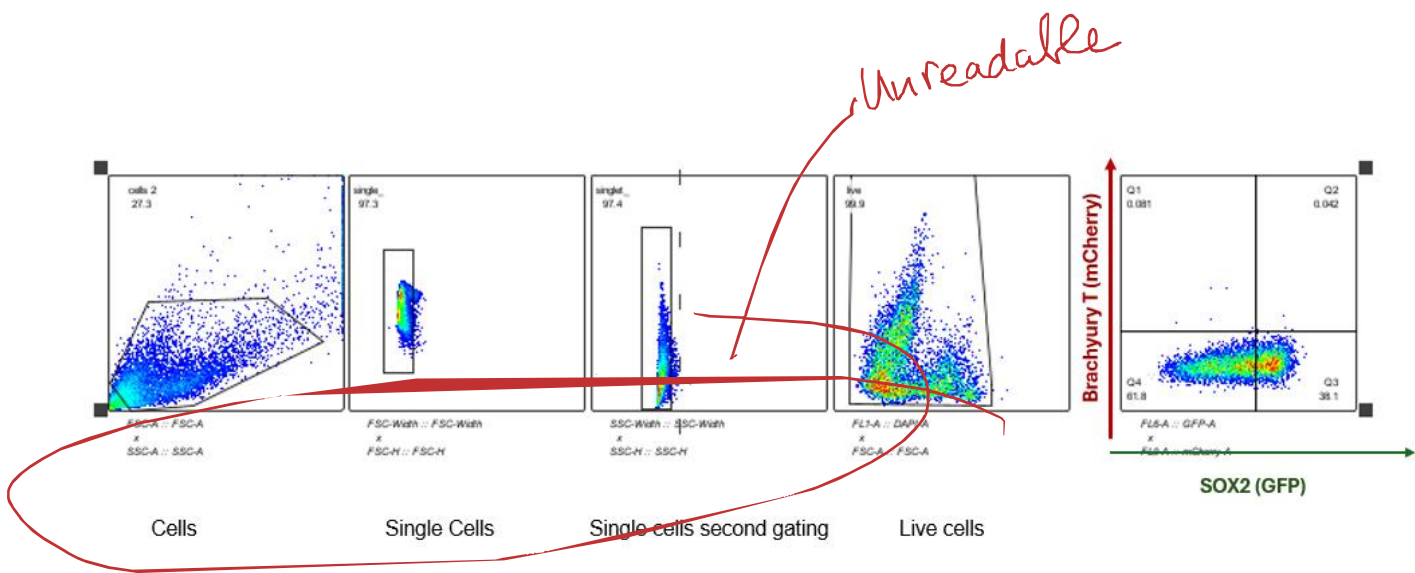
Supplementary Figure 1 | Localization of the lncRNA: RMST, C13Rik, and 913Rik. The UMAP plots, annotated to show distinct ectoderm and mesoderm regions, were generated from VASA sequencing data²⁶. The plots illustrate the expression patterns of three lncRNAs in mouse gastruloids (left) and day 5 (E5) mouse embryos (right). **a** RMST is predominantly localized to the ectoderm in both in both *in vitro* and *in vivo* datasets. **b** C13Rik exhibits similar localization as RMST. **c** 913Rik is primarily localized in the mesoderm in both data sets.

Target	Forward Primer Sequence (5'→3')	Reverse Primer Sequence (5'→3')
Gapdh	GGCAAATTCAACGGCACAGT	GTCTCGCTCCTGGAAGATGG
SOX1	TGGACTTCAACCGCAACTTAC	TCTCTTGGGTCCCTTTGACAT
PAX6	GCAGATGCAAAAGTCCAGGTG	CAGGTTGCGAAGAACTCTGTTT
Brachyury (T)	GCTTCAAGGAGCTAACTAACGAG	CCAGCAAGAAAGAGTACATGGC
MEOX1	GAAACCCCCACTCAGAAGATAGC	TCGTTGAAGATTCGCTCAGTC
TBX6	ATGTACCATCCACGAGAGTTGT	GGTAGCGGTAACCCTCTGTC
HPRT	CTGGTGAAAAGGACCTCTCGAAG	CCAGTTTCACTAATGACACAAACG

Supplementary Table 1 | RT-qPCR primers used in this study.

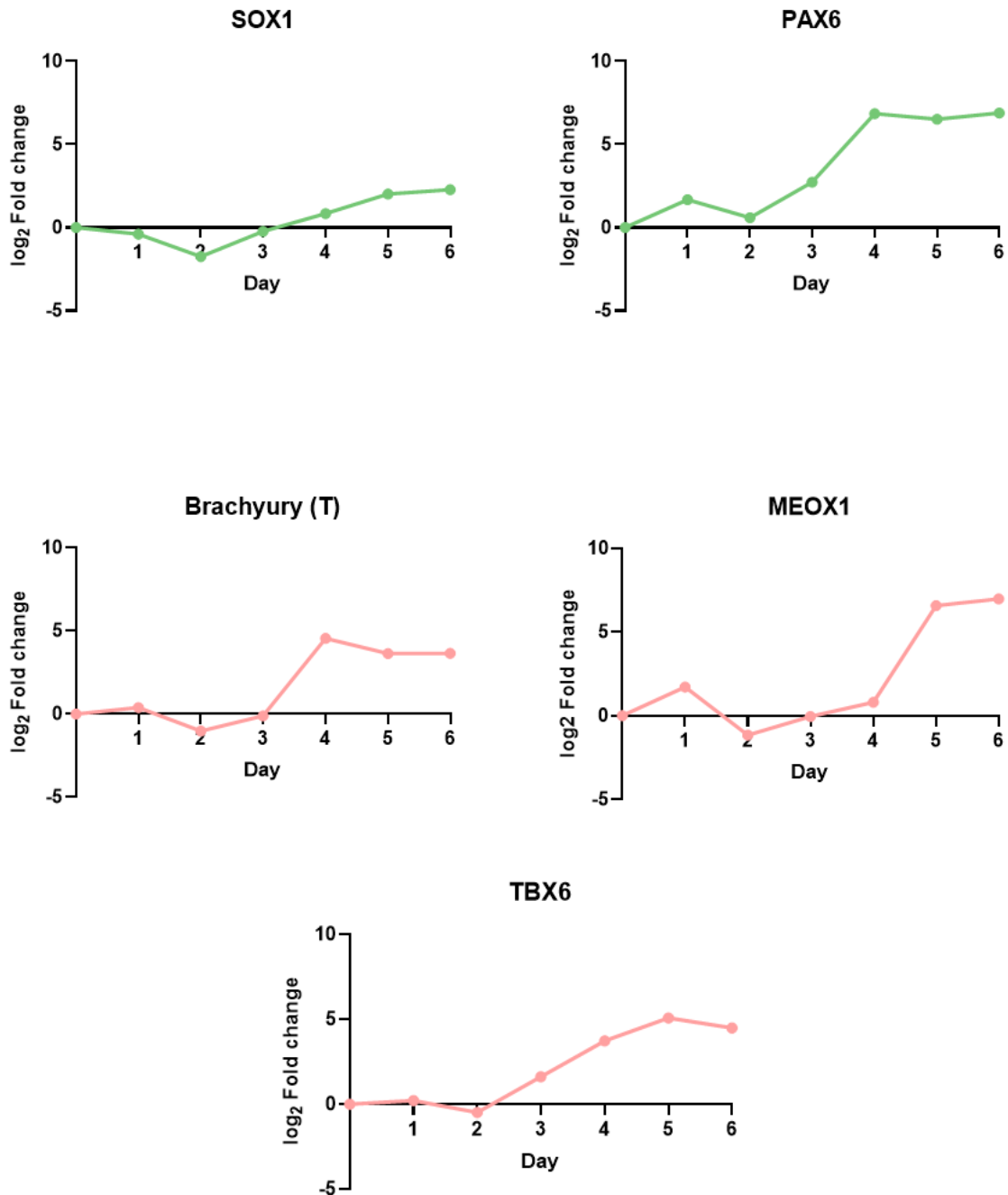
Target	Sequence
RMST 1	AAGTCTCCACGGTTCTATGG
RMST 2	CTTGGCGGTTGTACTCTTCA
RMST 3	TGGTGTCTTTCGTTGGCTTC
C13Rik 1	CATCAGGGACTTCTCGGATG
C13Rik 2	GAGGGTACAATCAGGTTTCC
C13Rik 3	GCAGTGTATGGCAGAAAGGA
913Rik 1	CATCACCGACATCAGGCAAT
913Rik 2	AGGACGACGTTTATCTGTGG
913Rik 3	AGCGCCATAGCTGAGTAGTA
Positive control (HPRT)	GCGACAATCTACCAGAGGGT
Negative control (random sequence)	CCTATAGGACTATCCAGGAA

Supplementary Table 2 | ASOs sequences targeting RMST, C13Rik, 913Rik, HPRT (positive control), and a random sequence (negative control). For each lncRNA, three ASOs were designed and synthesized with 2'-O-methyl (2'-OMe) modifications by Integrated DNA Technologies (IDT).

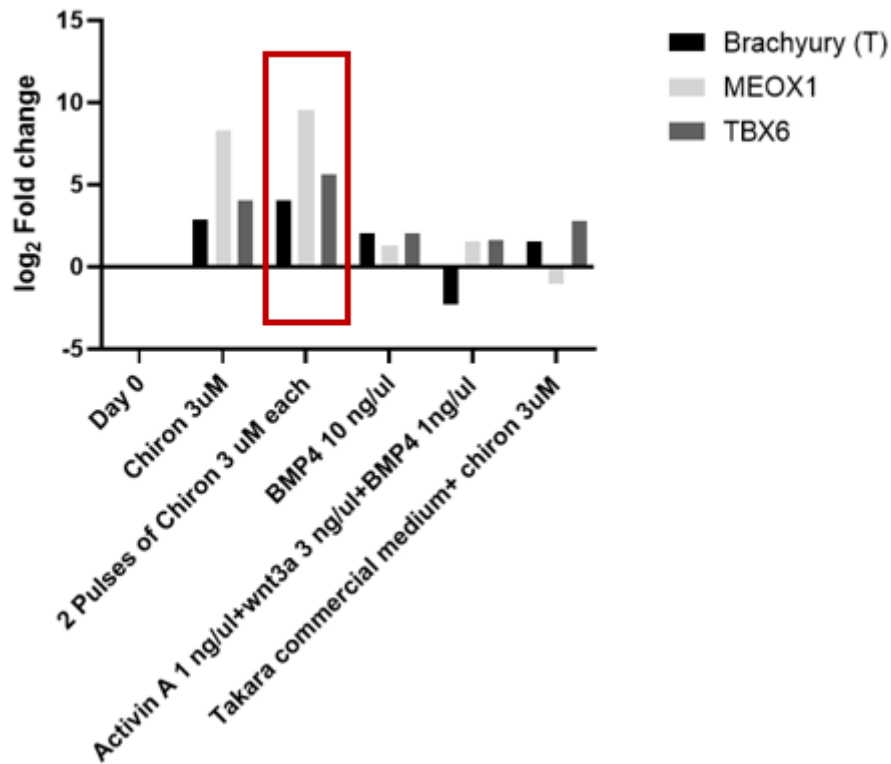


Supplementary Figure 2 | Representative flow cytometry plots explaining the gating strategy for isolating live single cells expressing SOX2 (GFP) and Brachyury T (mCherry).

Elaborate a bit more?



Supplementary Figure 3 | Log₂ fold change in expression of ectodermal markers (SOX1, PAX6) and mesodermal markers (Brachyury (T), MEOX1, TBX6) during ectoderm differentiation. The homemade differentiation medium initially tested, based on established protocols²⁵, resulted in increased expression of mesodermal markers Brachyury (T), MEOX1, and TBX6 after day 3 of differentiation.



Supplementary Figure 4 | Log₂ fold change in expression of mesoderm-specific markers (Brachyury (T), MEOX1, and TBX6) under different mesoderm differentiation protocols. To establish an efficient mesoderm differentiation method, several published protocols were tested. Among these, treatment with 3 μ M Chiron on days 2 and 3 of differentiation yielded the most consistent results, as shown by RT-qPCR analysis of the mesoderm-specific markers Brachyury (T), MEOX1, and TBX6. This protocol was selected for further optimization.